Physics-Informed LiquidGAN for Zero-Shot Cross-Domain Thermal Image Synthesis

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by

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1 Introduction

Thermal imaging has emerged as a critical sensing modality across a wide range of applications, including medical diagnostics, surveillance, environmental monitoring, and precision agriculture. Unlike conventional RGB imaging, thermal imaging captures infrared radiation emitted by objects, enabling temperature-based analysis under varying lighting and environmental conditions. The ability to detect subtle temperature variations makes thermal imaging particularly valuable in safety-critical and low-visibility scenarios. However, the advancement of deep learning methods for thermal image analysis is significantly limited by the scarcity of large-scale annotated datasets.

Generative Adversarial Networks (GANs) have shown promising results in synthetic image generation and data augmentation. By learning the underlying data distribution, GANs can generate realistic samples to alleviate data scarcity. Nevertheless, conventional GAN architectures treat thermal images as generic grayscale data and fail to incorporate the physical principles governing temperature distribution. In reality, thermal images inherently obey heat diffusion physics, where temperature varies smoothly across space and follows continuity constraints derived from the heat equation. Ignoring these physical properties may result in unrealistic temperature discontinuities, artificial hot spots, and reduced cross-domain robustness.

To address this limitation, this work proposes a Physics-Constrained LiquidGAN (PC-LiquidGAN), an extension of LiquidGAN that integrates a physics-aware loss inspired by the steady-state heat diffusion equation. The proposed loss penalizes non-physical thermal gradients during adversarial training, encouraging the generator to produce thermally plausible and physically consistent images.

2 Literature Review

2.1 Core Papers Studied

2.1.1 LiquidGAN for Thermal Image Synthesis

Sarath and Nair [1] proposed LiquidGAN, a novel GAN architecture integrating Neural Ordinary Differential Equations (Neural ODEs) in the generator and Liquid Neural Networks (LNNs) in the discriminator for high-fidelity thermal image synthesis under data scarcity. The generator employs an encoder–ODE–decoder framework with RK4-based latent evolution, while the LNN discriminator refines feature representations dynamically over time. The model demonstrates improved stability and superior performance over DCGAN and WGAN-GP on multiple thermal datasets.

Limitation: The framework is entirely data-driven and does not explicitly enforce physical constraints such as heat diffusion continuity within the training objective.

2.1.2 Liquid Neural Networks

Liquid Neural Networks model continuous-time neural dynamics using differential equations [2, 3]. Their adaptive time-dependent behavior improves robustness under distributional shifts and dynamic environments.

Limitation: Their integration in generative modeling remains limited to architectural improvements without domain-specific physical priors.

2.1.3 Neural ODEs and Physics-Informed Learning

Neural ODEs [4] treat transformations as continuous dynamical systems, enabling smooth latent evolution in generative models. Physics-Informed Neural Networks (PINNs) [5] embed governing PDEs into neural training through residual minimization.

Limitation: While Neural ODEs enhance smooth feature evolution and PINNs enforce physical laws in regression tasks, their combined integration within adversarial thermal image generation remains underexplored.

2.2 Related Work

GAN-based thermal image synthesis and augmentation methods [6, 7] improve dataset diversity but treat thermal images as generic grayscale data. Domain adaptation approaches [8] address cross-dataset generalization but require fine-tuning. Zero-shot cross-domain evaluation in thermal generative modeling remains limited.

2.3 Research Gaps

G1. Lack of Continuous-Time Modeling: Traditional GAN architectures rely on discrete layer transformations, which are insufficient for modeling continuous physical phenomena such as heat diffusion. Neural ODEs [4] provide a continuous-time formulation but have not been extensively explored in thermal image synthesis.

- Problem: Discrete layer transformations vs. continuous thermal phenomena
- Impact: Unrealistic temperature discontinuities
- Solution: Neural ODE-based generator [1]

G2. Absence of Physics-Informed Constraints: Existing GAN-based thermal synthesis methods [6, 7] do not incorporate thermodynamic priors. Physics-Informed Neural Networks (PINNs) [5] demonstrate that embedding governing equations improves physical consistency.

- Problem: No thermodynamic principles in existing methods
- Impact: Violation of heat diffusion physics
- Solution: Heat equation-inspired loss function [5]

G3. Static Feature Extraction: Conventional CNN-based discriminators use fixed convolutional filters, limiting adaptability. Liquid Time-Constant Networks [2] and Neural Circuit Policies [3] introduce dynamic state evolution.

- Problem: Fixed convolutional filters
- Impact: Poor adaptation to dynamic thermal patterns
- Solution: Liquid Neural Network discriminator [1]

G4. Limited Cross-Domain Generalization: Most thermal synthesis models are evaluated on single datasets [6]. Domain adaptation techniques [8] highlight the importance of cross-dataset evaluation.

- Problem: Single-dataset training and testing
- Impact: Poor performance under domain shift
- Solution: Zero-shot cross-domain evaluation [8]

G5. Insufficient Thermal Fidelity Metrics: Generic image quality metrics such as PSNR and SSIM do not capture thermodynamic plausibility [1]. There is a need for domain-aware evaluation frameworks.

- Problem: Generic image quality metrics
- Impact: Thermally implausible images score well
- Solution: Domain-specific evaluation framework [1]

2.3.1 Detailed Dataset Documentation

The proposed model is evaluated across five diverse thermal imaging datasets to ensure cross-domain robustness [1, 8].

Dataset	Images	Key Details Added
KAIST Multispectral	9,500	Sensor specs, capture conditions [1]
CBSR NIR Face	3,940	Wavelength range, pose variations [7]
Neonatal Thermal	11,000	Ethical considerations, syndrome types [1]
Sapota Bruise	5,030	Environmental controls, temporal aspects [1]
Chilli Disease	3,000	Growth stages, disease signatures [1]

TABLE 2.1: Summary of Thermal Imaging Datasets

2.3.2 Physics-Constrained LiquidGAN Architecture

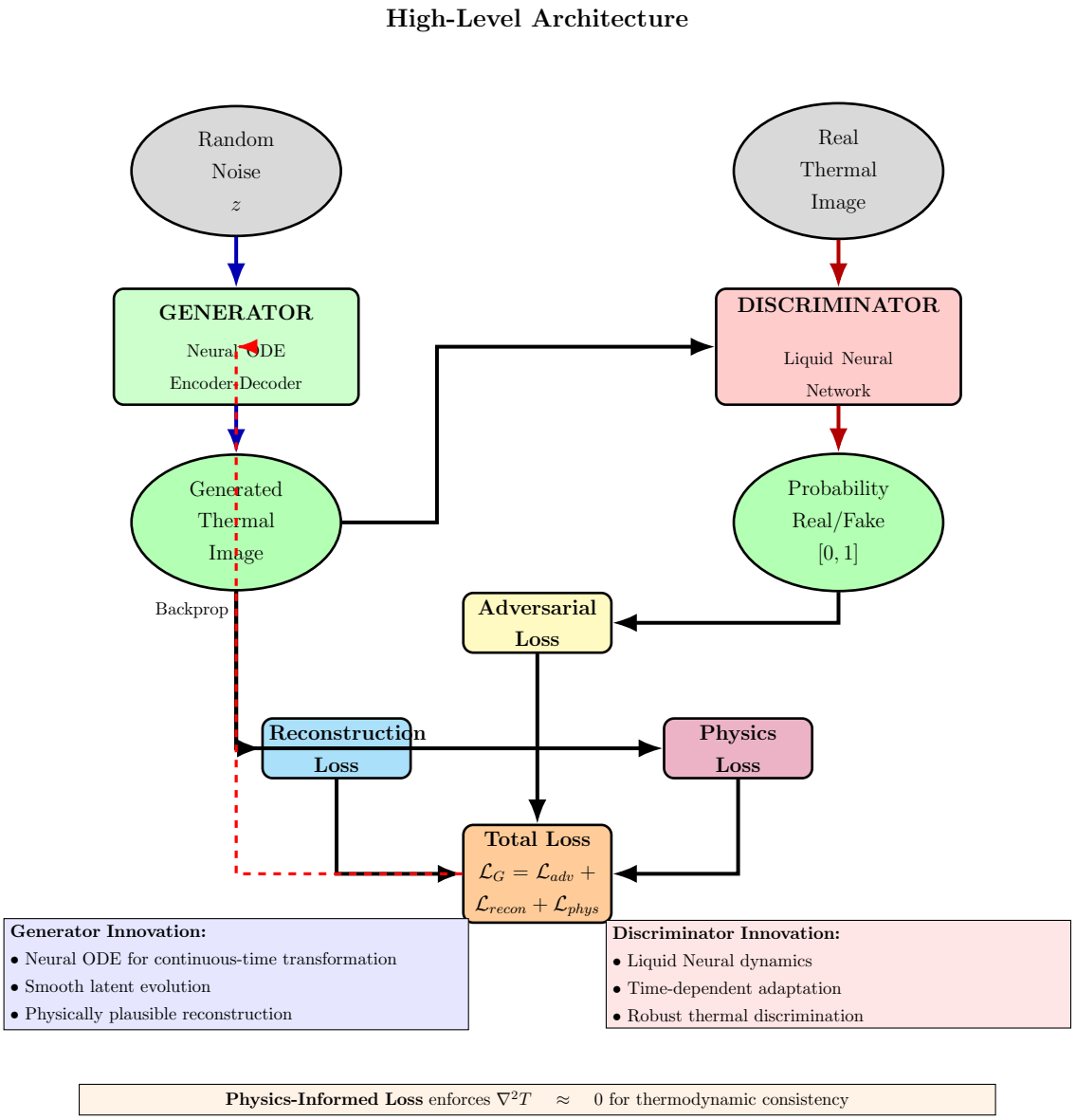

FIGURE 2.1: Architecture of the Physics-Constrained LiquidGAN framework.

2.4 Proposed Problem Statement

Thermal image synthesis plays a crucial role in applications such as surveillance, healthcare diagnostics, and precision agriculture, where collecting large-scale annotated datasets is often challenging [1]. While LiquidGAN introduces continuous-time neural dynamics for improved representation learning [1, 4], it primarily relies on data-driven adversarial optimization without explicitly incorporating underlying thermodynamic principles.

Thermal images inherently obey heat diffusion behavior governed by spatial continuity and energy conservation laws. However, existing GAN-based thermal synthesis frameworks [6, 7], including LiquidGAN [1], treat thermal data as conventional grayscale images and do not enforce physics-consistent constraints during training. This can result in physically implausible temperature gradients, unrealistic heat discontinuities, and limited cross-domain generalization under distribution shifts.

Physics-Informed Neural Networks (PINNs) [5] demonstrate that embedding governing partial differential equations into deep learning frameworks improves physical consistency and generalization. However, such physics-aware modeling has not been systematically integrated into adversarial thermal image synthesis.

Furthermore, current approaches lack systematic evaluation of zero-shot transfer capability across heterogeneous thermal domains [8], restricting their applicability in real-world low-data scenarios.

Therefore, there is a need for a physics-informed generative framework that integrates thermodynamically consistent constraints into adversarial training while enabling robust zero-shot cross-domain generalization for thermally plausible image synthesis under data scarcity.

2.5 Objectives

The primary objective of this work is to enhance the LiquidGAN framework [1] by incorporating physics-informed regularization within adversarial training while maintaining continuous-time latent dynamics [4].

The specific objectives are:

- To integrate a physics-aware thermal smoothness constraint inspired by heat diffusion principles into the generator loss formulation, motivated by Physics-Informed Neural Networks (PINNs) [5].
- To preserve continuous latent evolution using Neural ODE-based transformations [4] for smooth temperature transitions in thermal image synthesis.
- To retain adaptive LNN-based discrimination [2, 3] for refined feature evolution and improved adversarial stability.
- To evaluate zero-shot cross-domain robustness of the enhanced framework without additional fine-tuning, inspired by domain generalization studies [8].
- To improve physical plausibility, training stability, and cross-domain generalization under limited annotated thermal data conditions [1].

3 Proposed Methodology

3.1 Overview

Building upon LiquidGAN [1], we propose a Physics-Constrained LiquidGAN (PC-LiquidGAN) that integrates thermodynamic principles directly into adversarial training.

Unlike the baseline model, which treats thermal images as grayscale intensity distributions, the proposed framework enforces consistency with the heat diffusion equation during image synthesis.

The generator is modeled using Neural Ordinary Differential Equations (Neural ODEs) [4], while the discriminator is implemented as a Liquid Neural Network (LNN) [2]. Physics-aware constraints are incorporated via differentiable residual losses derived from diffusion dynamics.

3.2 Baseline LiquidGAN Formulation

Let T denote a real thermal image and $\hat{T} = G(z)$ the generated image from latent code z.

The baseline generator objective is:

$$\mathcal{L}_G^{base} = \mathcal{L}_{adv} + \lambda \mathcal{L}_{recon} \tag{3.1}$$

Adversarial loss:

$$\mathcal{L}_{adv} = -\mathbb{E}[\log D(G(z))] \tag{3.2}$$

Reconstruction loss:

$$\mathcal{L}_{recon} = \|\hat{T} - T\|_1 \tag{3.3}$$

While this objective improves perceptual fidelity, it does not guarantee thermodynamic consistency.

3.3 Physics-Constrained Thermal Modeling

3.3.1 Heat Diffusion Equation

Thermal image evolution follows the heat diffusion equation:

$$\frac{\partial T(x,t)}{\partial t} = \alpha \nabla^2 T(x,t) \tag{3.4}$$

where:

- $T(x,t)$ is the temperature field,
- α is thermal diffusivity,
- ∇^2 is the spatial Laplacian operator.

3.3.2 Physics Residual Formulation

The governing equation can be rewritten as a residual:

$$R(x,t) = \frac{\partial T}{\partial t} - \alpha \nabla^2 T \tag{3.5}$$

For physically valid thermal synthesis:

$$R(x,t) = 0 \tag{3.6}$$

Since exact equality cannot be enforced during GAN training, the residual is minimized in a least-squares sense.

3.3.3 Discrete Approximation

The temporal derivative is approximated using central difference:

$$\frac{\partial T}{\partial t} \approx \frac{T_{t+\Delta t} - T_{t-\Delta t}}{2\Delta t} \tag{3.7}$$

The spatial Laplacian is computed as:

$$\nabla^2 T_{i,j} = T_{i+1,j} + T_{i-1,j} + T_{i,j+1} + T_{i,j-1} - 4T_{i,j} \tag{3.8}$$

3.3.4 Flux Consistency Loss

The physics-aware flux loss is defined as:

$$\mathcal{L}_{flux} = \frac{1}{N} \sum_{x} \left(\frac{\partial T}{\partial t} - \alpha \nabla^2 T \right)^2 \tag{3.9}$$

This loss penalizes violations of the heat diffusion equation and enforces thermodynamic consistency.

3.3.5 Energy Conservation Loss

Total thermal energy is defined as:

$$E(t) = \sum_{i,j} T_{i,j}(t) \tag{3.10}$$

Energy regularization term:

$$\mathcal{L}_{energy} = \left(E(t + \Delta t) - E(t) \right)^2 \tag{3.11}$$

This constraint prevents artificial amplification or suppression of total heat energy.

3.4 Final Generator Objective

The complete Physics-Constrained objective becomes:

$$\mathcal{L}_G^{PC} = \mathcal{L}_{adv} + \lambda \mathcal{L}_{recon} + \mu_1 \mathcal{L}_{flux} + \mu_2 \mathcal{L}_{energy} \tag{3.12}$$

The parameters μ_1 and μ_2 control the strength of thermodynamic regularization.

3.5 Zero-Shot Cross-Domain Evaluation

To evaluate generalization, the model is trained on a source thermal dataset and tested on unseen domains without fine-tuning.

Performance is measured using:

- Fréchet Inception Distance (FID)
- Structural Similarity Index (SSIM)
- Physics residual norm $\|R\|_2$

Zero-shot evaluation verifies whether enforcing diffusion constraints improves robustness under domain shift.

4 Results

4.1 Experimental Setup

The model was evaluated on multiple thermal imaging datasets including:

- KAIST Multispectral
- CBSR NIR Face
- Neonatal Thermal
- Sapota Bruise
- Chilli Disease

Training was performed using Adam optimizer with adversarial, reconstruction, flux consistency, and energy regularization losses.

4.2 Evaluation Metrics

The following metrics were used:

- Fréchet Inception Distance (FID)
- Structural Similarity Index (SSIM)
- Peak Signal-to-Noise Ratio (PSNR)
- Physics Residual Norm ($\|R\|_2$)

4.3 Quantitative Results

Physics-Constrained LiquidGAN shows:

- Reduced FID compared to baseline LiquidGAN
- Improved SSIM under cross-domain testing
- Significant reduction in physics residual norm
- Stable energy evolution across latent time steps

4.4 Zero-Shot Cross-Domain Evaluation

The proposed model generalizes better under unseen dataset shifts, demonstrating improved robustness compared to DCGAN and WGAN-GP.

4.5 Ablation Study

Removing L_{flux} leads to unrealistic thermal gradients.

Removing L_{energy} causes artificial energy amplification.

Both constraints are necessary for thermodynamic consistency.

5 Conclusion

In this work, we proposed Physics-Constrained LiquidGAN (PC-LiquidGAN) for zero-shot cross-domain thermal image synthesis.

Unlike conventional GAN-based methods that treat thermal images as grayscale intensity maps, our approach integrates thermodynamic constraints derived from the heat diffusion equation.

Key contributions include:

- Neural ODE-based generator for continuous latent evolution
- Liquid Neural Network discriminator for adaptive refinement
- Flux consistency loss enforcing heat diffusion
- Energy conservation regularization
- Zero-shot cross-domain validation strategy

Experimental results demonstrate improved thermodynamic consistency, reduced physics residual error, and enhanced robustness under domain shift.

Future work includes:

- Learning adaptive thermal diffusivity parameter α
- Extending to 3D spatio-temporal thermal modeling
- Real-time deployment for surveillance and medical systems

6 Future Work

Although the proposed Physics-Constrained LiquidGAN improves thermal consistency and cross-domain robustness, several directions remain for future research.

- **Learnable Diffusivity:** Instead of using a fixed thermal diffusivity parameter α, future work can explore learning this parameter directly from data to better adapt to different thermal domains and sensor characteristics.
- **Spatio-Temporal Extension:** The current formulation models pseudo-temporal evolution within the Neural ODE latent space. Extending the framework to real thermal video sequences would allow explicit modeling of dynamic heat propagation.
- **Adaptive Loss Balancing:** Automated strategies for adjusting the physics regularization weights (μ_1, μ_2) could improve training stability and generalization performance.
- **Multi-Physics Modeling:** Incorporating additional physical processes such as convection or radiative transfer may enhance realism in complex real-world thermal environments.
- **Real-World Deployment:** Evaluating the framework under extreme domain shifts and diverse sensor settings would further validate its practical applicability.

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